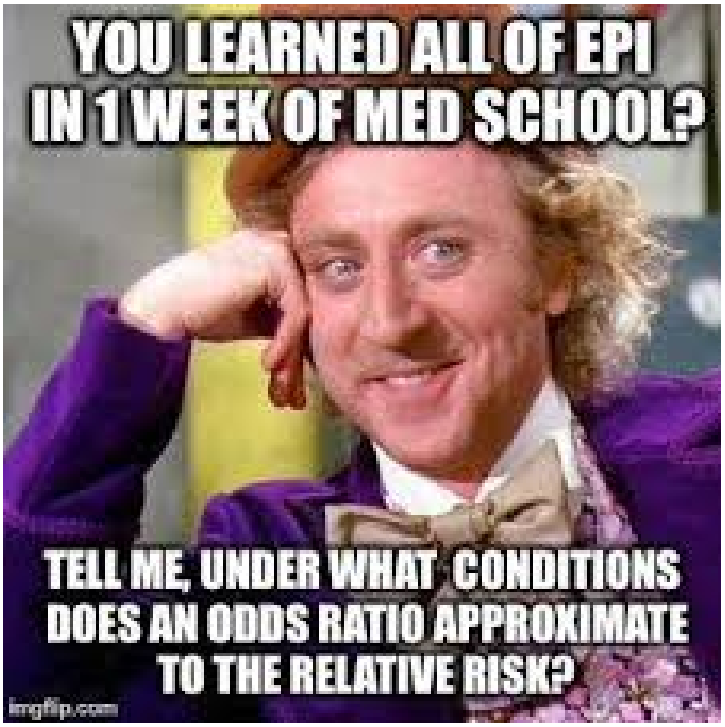
Disease Detectives

Captain's Summer Exchange 2025



2025-2026 Season Tryouts Test

Hillsborough High School

Division C

Written by Anya Verma

Name: _____

Score: _____/186 points

Instructions:

1. You will have 50 minutes to complete this test
2. You may bring one 8.5" x 11" sheet of paper, which may be in a sheet protector or laminated, with information on both sides from any source
3. You are allowed two stand-alone non-programmable, non-graphing calculators (Class II)
4. Points are denoted in the square brackets ([]) next to the question
5. Partial credit may be awarded to open-ended questions
6. Tiebreakers will be denoted with "[TB #]" and scored in the following order: **34, 23e, 43, 44, 45a**

Points and Questions Breakdown

Vocabulary	Page 3-4	____/20 points
Background & Surveillance	Page 5-8	___/60 points
Outbreak Investigation	Page 9-14	___/55 points
Miscellaneous	Page 15	___/51 points

GOOD LUCK!!

Section 1: Vocabulary ___/20 points

Match each term to its best description. Terms may be used only once, multiple times, or not at all.

A. Pathogenicity	E. Iatrogenic	I. Antigenic Drift	M: Analysis of Variance (ANOVA)	Q: Positive Chemotaxis
B. Virulence	F. Fulminant	J. Prevalence	N: Convalescence	R: Negative Chemotaxis
C. Morbidity	G. Sufficient Cause	K. Secular Trend	O: Anthroponosis	S: Incidence
D. Nosocomial	H. Antigenic Shift	L. Iceberg Phenomenon	P: Serotype	T: None of the Above

1. [1] Compares observed and expected frequencies of events, determining if a relationship exists or if the differences are due to chance.

   Answer: _____
2. [1] Disease acquired from a hospital.

   Answer: _____
3. [1] The period of time when individuals are recovering from a disease.

   Answer: _____
4. [1] A minimum set of requirements, that if present in an individual, will result in the disease.

   Answer: _____
5. [1] The proportion of persons infected, after exposure to a causative agent, who then develop clinical disease.

   Answer: _____
6. [1] The attribution of human characteristics to another entity.

   Answer: _____
7. [1] The number of affected individuals at any given point in time.

   Answer: _____

8. Can be broken into two main types: one-way & two-way.

   Answer: _____

9. [1] Severe and sudden in onset.

   Answer: _____

10. [1] The number of new cases that occurred during a specific time.

    Answer: _____

11. [1] Changes over a relatively long period of time.

    Answer: _____

12. [1] Minor genetic changes that result in slightly different strains of the same virus.

    Answer: _____

13. [1] Any departure (subjective or objective) from a state of well-being.

    Answer: _____

14. [1] A way for grouping microorganisms together based on their similarities and differences.

    Answer: _____

15. [1] Encompasses disease spread from human-to-human and human-to-animal

    Answer: _____

16. [1] When a microorganism moves to an area of high concentration of a specific chemical.

    Answer: _____

17. [1] Illness acquired through medical treatment.

    Answer: _____

18. [1] The proportion of persons with clinical disease, who after becoming infected, become severely ill or die.

    Answer: _____

19. [1] Assesses the statistical means between two or more different groups.

    Answer: _____

20. [1] When a trend appears in smaller groups but disappears when the groups are combined.

    Answer: _____

Section 2: Background and Surveillance

____/60 points

Follow the directions provided for each question set.

21. [10] What are the types of prevention? Give a brief description of each.

22. Categorize the following statements into a clinical (C) or public health (P) approach:
    a. [1] Dr. Grey takes Steve's blood pressure and puts it into his chart.

       Answer: _____
    b. [1] Dr. Bailey sets up multiple COVID-19 testing clinics throughout Seattle to analyze disease spread.

       Answer:_____
    c. [1] Dr. Karev and Dr. Stevens pass around pamphlets on the dangers of STDs after several staff members at Seattle Grace Hospital have contracted syphilis.

       Answer: _____
    d. [1] After observing the rising rates of death among patients who have undergone spinal repair surgery, Dr. Shepherd launches a program to track his patients' health post-surgery where they are offered highly personalized advice.

       Answer: _____

23. Name the person based on their description:

    a. [2] Was immune to typhoid, but spread it to 50 other people due to a lack of hygiene.

       Answer: _____

    b. [1] Discovered penicillin.

       Answer: _____

    c. [1] Developed the oral polio vaccine.

       Answer: _____

    d. [1] Father of immunology.

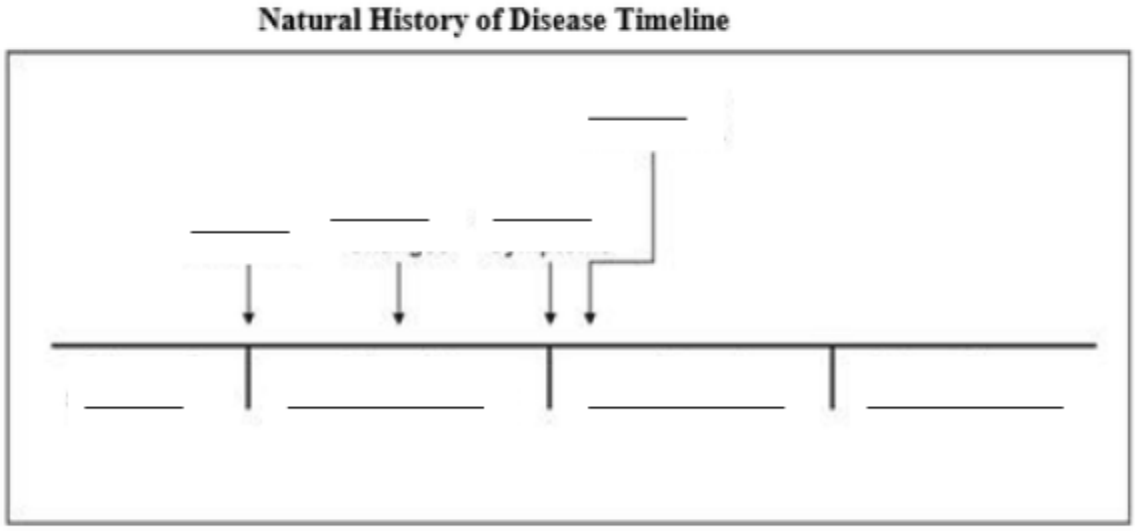
       Answer: _____

    e. [2][TB **2**] Studied radiation induced illness among workers at the Hanford plutonium production plant.

       Answer: _____

24. [13] Name the 13 steps of outbreak investigation.

25. [8] Label the Natural History and Spectrum of Disease:



a. [1] Addison's husband has COVID-19 but she does not. What stage in the Natural History and Spectrum of Disease is she in?

   Answer: _____

b. [1] Eobard Thawne has infected Iris's coffee with a poisonous spore, which she drank this morning. Iris is not showing any symptoms, however. What stage in the Natural History and Spectrum of Disease is she in?

   Answer: _____

c. [1] Leonard Snart recovers from a flu which he acquired from his activities in the cold. What stage in the Natural History and Spectrum of Disease is he in?

   Answer: _____

26. [9] What is surveillance? What are the 5 steps of surveillance?

27. Use the following types of surveillance to describe the situations.

    Passive Active Sentinel Syndromic

a. [2] All hospitals in Central City report all cases of anthrax to the local health department.

   Answer: _____

b. [2] Starling City's health department notices that there has been a rise in school absences at the peak of flu season, and increases availability of flu vaccines.

   Answer: _____

c. [2] The CDC asks the local health departments in all cities bordering and including National City to provide statistics on who was infected with Kryptoflu.

   Answer: _____

Section 3: Outbreak Investigation ___/55 points

Use the provided information to answer the questions.

In March of 2017, a number of Central City citizens attended S.T.A.R. Labs for a party to celebrate Barry Allen's successful defeat of Savitar. After the party, however, a number of people called in sick the next day, reporting symptoms such as watery diarrhea, nausea, and vomiting. Dr. Caitlin Snow was not at the party but has been assigned to investigate the outbreak, and she needs your help.

28. [4] Write a case definition for this outbreak.

Dr. Snow believes that everyone has been afflicted with a foodborne disease. She has given you the task to contact everyone who attended the party to figure out what they ate, so that you can pinpoint the specific culprit that is responsible for the sickness.

29. [4] Name the four fundamental steps to food safety:

30. [6] What kind of case study are you conducting (if necessary, be specific)? How do you know? Give one pro and one cons.

31. [1] You ask Barry Allen, one of the people who got sick, what he ate during the party. He responds, "Uhh, I don't really remember, I was eating everything so fast that it kind of all just flew past my head." What kind of bias is this?

	Disease	No Disease
Spaghetti	22	36
Caviar	52	34
Steak	12	40
Cobb Salad	35	45
Chicken Tortilla Soup	14	20

32. [11] The table above shows the foods served at the party, as well as the disease outcome for those who ate the food. State and calculate the appropriate measure of risk for each food to the thousandth. Show all work.

33. [4] What was the most likely culprit for the outbreak? Explain in terms of the measure of risk you calculated.

34. [TB, 1][3] Speaking of measures of risk, the meme on the front cover gives a good example of two well-known ones. Can you answer Willy Wonka's compelling question: "Under what conditions does

an odds ratio approximate relative risk?" What is this called?

Dr. Snow comes to the appalling realization that Slade Wilson, a former member of the Australian Intelligence Service, has actually engineered a pathogen to target one of the party's attendees, Oliver Queen. Before the food was delivered to the party, Slade had infused one of Oliver's favorite foods with his pathogen.

You just get news that Slade has unleashed the pathogen to all of Central City's residents. Your amazing colleague, Cisco Ramon, has devised a test he calls "Vibe-X" that will help you determine who is sick from the pathogen and who is not, to aid your research in creating a vaccine. Vibe-X gives results as either positive or negative, and you decided to test how good it was, with the following results:

	Sick	Healthy
Positive	92	20
Negative	12	112

35. [3] Define sensitivity and calculate the sensitivity of Vibe-X.

36. [3] Define specificity and calculate the specificity of Vibe-X.

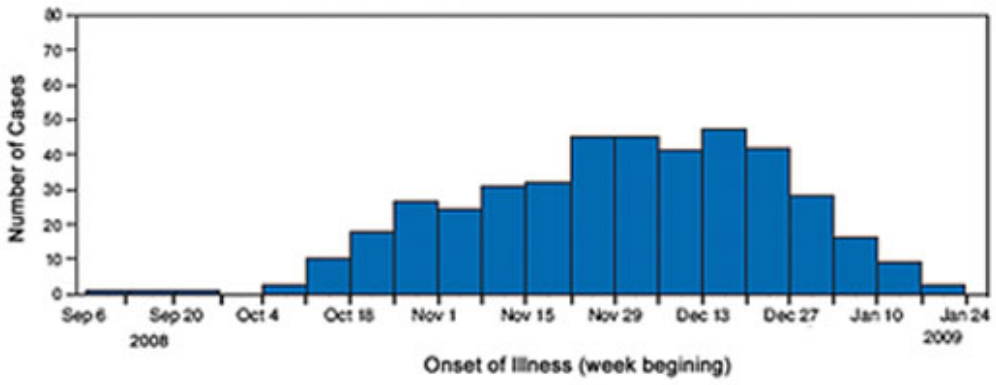
37. [3] Define PPV and calculate the PPV of Vibe-X.

38. [3] Define NPV and calculate the NPV of Vibe-X.

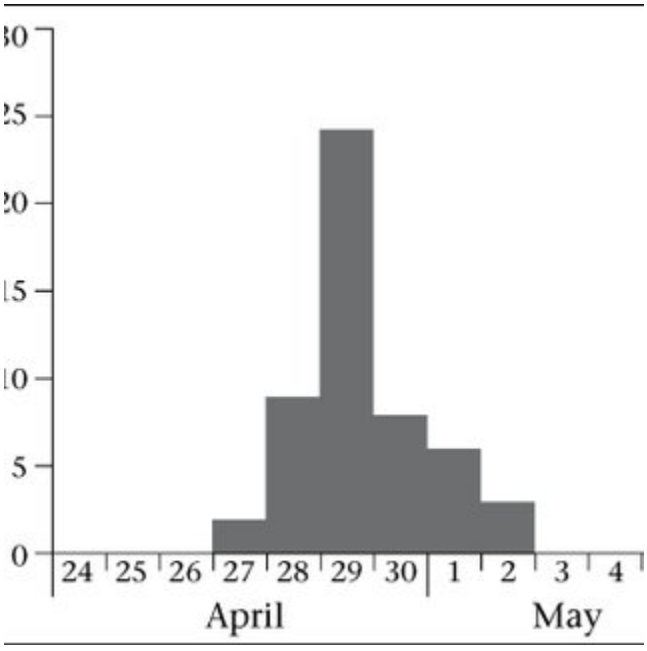
You and Dr. Snow quickly get to work making a vaccine. However, Lex Luthor shows up and decides that you can not go on further with your efforts until you identify a series of epi curves. Lucky for you, you've been catching up on your epi curve research and know just how to dodge these questions:

39. [2] What is an epi curve?

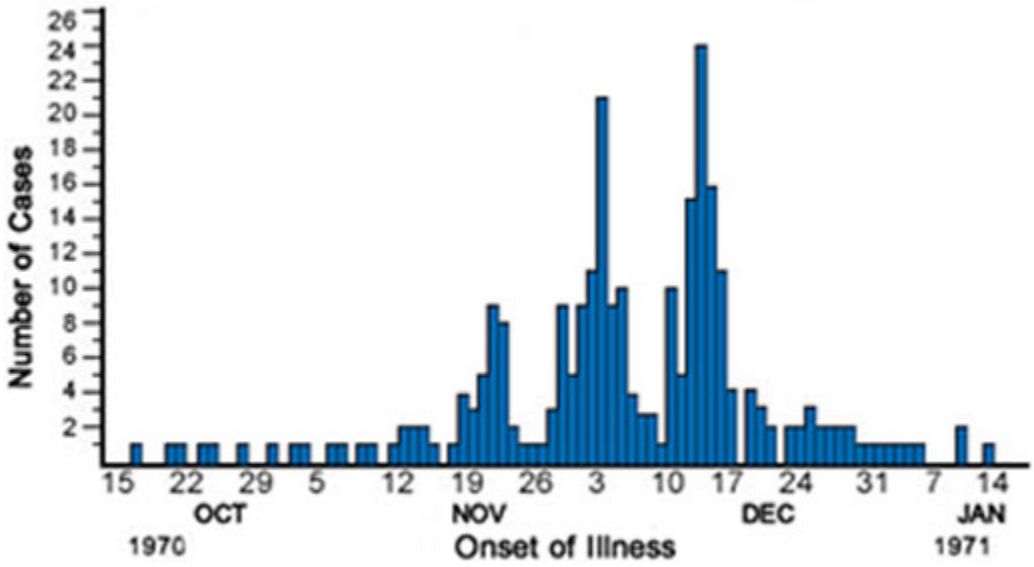
40. [1] Identify the epi curve:



41. [1] Identify the epi curve:



42. [1] Identify the epi curve:



You and Dr. Snow administer the vaccine to the citizens of the city. As their symptoms and disease subside, you and all of the S.T.A.R. Labs personnel have another party for a job well done—this time, making sure that no one can tamper with the food.

You create a vaccine called Aura-X and decide to run a clinical trial with the people of Central City with the following results:

	Positive	Negative
Vaccinated	7	37
Unvaccinated	57	6

43. [2] Differentiate between vaccine efficacy and effectiveness.

44. [2] Calculate the Aura-X's effectiveness and express as a percent rounded to the hundredths.

45. [1] What is one flaw in this clinical trial? Hint: Think about the people tested.

You and Dr. Snow administer the vaccine to the citizens of the city. As their symptoms and disease subside, you and all of the S.T.A.R. Labs personnel have another party for a job well done—this time,

Section 4: Miscellaneous ____/51 points

46. [TB **3**] [8] What is the Framingham Study? Include information on findings, the study itself, and important dates.

47. [TB 4] [4] What are the next 2 countries where scientists are trying to eradicate polio?

48. What do they stand for?
    a. [TB 5][1] AAP:
    b. [1] NNDSS:
    c. [1] GPEI:
    d. [1] CDC:
    e. [1] HIPAA:

49. [10] Name all of Evan's Postulates:

50. The mean R_0 of COVID-19 is approximately 3.
    a. [2] What is an R_0 value?

    b. [2] What is the herd immunity threshold for COVID-19? Show work.

True or False

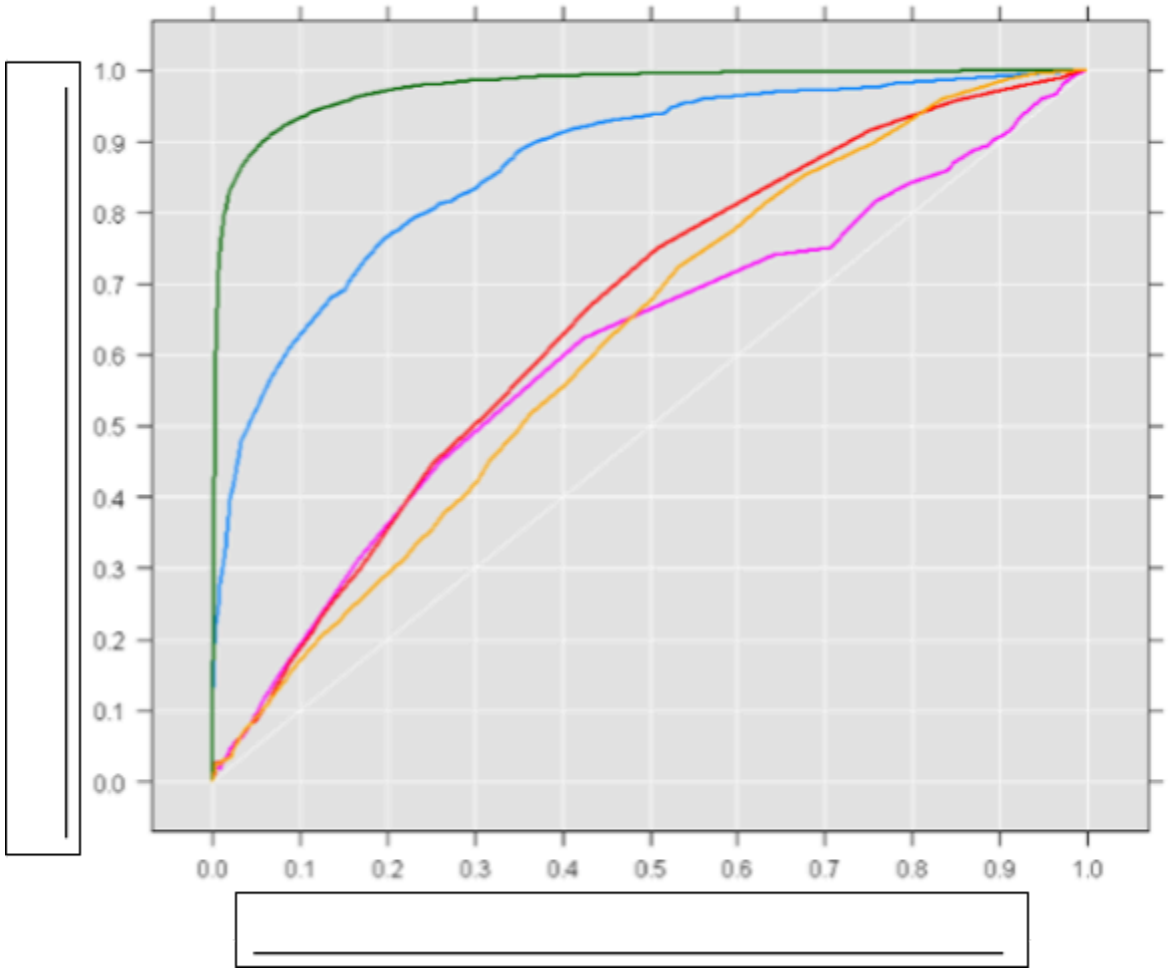
51. [1] Minamata disease is often called Dancing Cat Fever

52. [1] You can use antibiotics to treat shigatoxin-producing E. coli.

53. [1] Anthrax is a zoonotic disease coming from snakes.

54. [1] Mechanical transmission is involved when the disease does not go through biological transformation in a vector.

55. Look at the following curve:



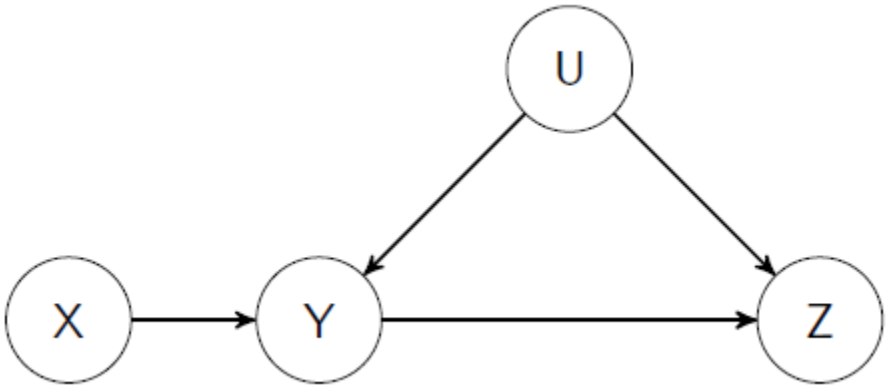
a. [1] What is the name of this kind of curve?

b. [2] Label the axes of the curve.

c. [1] In the context of epidemiology, name one thing that the curves could represent.

d. [1] Epidemiologists want to choose the curve that represents the most effective variable. Which curve should they choose?

56. Look at the following diagram:



a. [1] What is this diagram's name?

b. [1] What does the letter Z represent?

c. [1] What does the letter U represent?

d. [5] Create your own diagram below like the one above for a disease of your choosing. Nodes and arrows do not have to be the same.

57. [3] Name one breakthrough in public health, medicine, or epidemiology that you feel has revolutionized its field. Explain in detail what it is and how it works, and why you chose this.

End of Test

Congrats!!